Tasks for today

- Pull the assignment `ising_ensemble_avg_package.ipynb` from the QuantumSoftware GitHub repository
- Activate the virtual environment `molssi_best_practices` from earlier
- Create a Python package called montecarlo using cookiecutter
- Resource: Python package setup tutorial from earlier in the class will be helpful: https://education.molssi.org/python-package-best-practices/01-package-setup.html

- Add your BitString class to your package montecarlo
- Define a new class called IsingHamiltonian in your package

Steps to create your new package `montecarlo`

- Run:
  `cookiecutter gh:audreyfeldroy/cookiecutter-pypackage`
- Go through each prompt it asks you and enter your specifications.
- Go into the top-level directory of your new package:
  `cd montecarlo`
- Next, in order to import your package and use it from anywhere on your computer, you will need to perform a development installation. You will then be able to import your package into scripts in the same way you import `matplotlib` or `numpy`.
- Run:
  `pip install -e .`

- If this errors saying that the Python version is incompatible, check what version of Python is in your virtual environment (run `conda list`), go into the file pyproject.toml and change the line
  `requires-python = ">= 3.12"`
  to
  `requires-python = ">= 3.10"`
  or to any compatible version.

Steps to create classes in your package

- Pull up the file where you have your BitString class from earlier

- Copy the code into the __init__.py file in montecarlo/src/montecarlo/

- Let’s test whether it works!